

Reliable and Trustworthy Artificial Intelligence

Assignment #3

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1. Exploration of Marabou Resources

The Marabou GitHub repository provides sample models and verification examples under the resources/ directory, organized as follows:

Models (format: .nnet, .onnx)

- resources/nnet/ — ACAS Xu networks (aircraft collision avoidance). Multiple fully-connected networks in .nnet format, e.g. ACASXu_experimental_v2a_1_1.nnet. Architecture: 5 inputs, 5 outputs, 6 hidden layers of 50 neurons each.
- resources/onnx/ — Small MNIST classifiers and other benchmark networks in ONNX format.

Datasets / Input Specifications

- ACAS Xu uses normalized sensor inputs (intruder distance, angles, speeds) with explicit bound constraints defined in accompanying property files.
- MNIST examples use pixel values normalized to [0, 1].

Example Verification Queries

- ACAS Xu properties 1–10: safety properties such as "if the intruder is distant, the advisory should not be COC (clear-of-conflict)."
- Robustness queries on MNIST: L_∞ -perturbation queries similar to this assignment.

The dominant format is .nnet for ACAS Xu and .onnx for vision models. Marabou also supports .pb (TensorFlow) but most examples use the former two.

2. Model and Dataset

The dataset used in this project was MNIST handwritten digit classification. A 5,000-sample subset was selected for training using a stratified split with random seed 42, and pixel values were normalized to [0,1] using MinMaxScaler.

The model was a small fully-connected neural network trained using scikit-learn's MLPClassifier. Its architecture is summarized in Table 1.

Layer	Size	Activation
Input	784	—
Hidden 1	32	ReLU
Hidden 2	16	ReLU
Output	10	— (logits)

Table 1. Architecture of the MNIST MLP model

The model achieved 92.0% test accuracy after 30 training iterations, which was sufficient for verification while keeping the network small enough for Marabou to handle efficiently. The model was exported to ONNX with onnx.shape_inference.infer_shapes applied, which is necessary for Marabou's ONNX parser to correctly resolve intermediate tensor shapes.

This model was chosen because it is not included in Marabou's resources directory, is small enough for verification to complete in seconds per query, and represents a realistic classification task.

3. Verification Query

For a reference input x (correctly classified as digit c), the query asks:

Does there exist x' such that $\|x' - x\|_\infty \leq \varepsilon$ and the network classifies x' as some class $j \neq c$?

This is encoded as:

- Input bounds: $x_i - \varepsilon \leq x'_i \leq x_i + \varepsilon$, clipped to $[0, 1]$ per pixel
- Output constraint (negated property): $\text{output}[j] - \text{output}[c] \geq 0$

One query is issued per rival class j . UNSAT means no such adversarial input exists (provably robust); SAT returns a concrete counterexample.

4. Results

Digits 0, 1, 2 were verified at two perturbation radii:

Digit	ε	Result	SAT rival(s)	Time
0	0.01	UNSAT (robust)	—	0.44s
0	0.05	SAT	class 2	45.85s
1	0.01	UNSAT (robust)	—	0.53s
1	0.05	SAT	class 2, 3, 8	32.45s
2	0.01	UNSAT (robust)	—	0.34s
2	0.05	UNSAT (robust)	—	0.55s

Table 2. Verification results under different perturbation bounds

Interpretation: All three digits are provably robust at $\varepsilon = 0.01$ — no adversarial perturbation within a 1% pixel range can change the classification. At $\varepsilon = 0.05$, digits 0 and 1 become vulnerable. Notably, digit 2 remains robust even at $\varepsilon = 0.05$, likely due to its visually distinctive shape having a larger margin from other classes in the learned representation. The digit 0 vs. class 5 query took 45 seconds — significantly longer than others — likely reflecting increased SMT solver complexity before proving the property UNSAT.

The counterexample found for digit 0 at $\varepsilon = 0.05$ vs. class 2 was identified within 0.063s, indicating reduced robustness at larger perturbation levels.

5. Marabou: Strengths and Limitations

Strengths

- Provides formal guarantees: UNSAT results are proofs, not heuristic estimates.
- Handles ReLU networks exactly without approximation, unlike abstract interpretation methods.
- Python API (maraboupy) makes it easy to load ONNX models and programmatically set constraints.

Limitations

- Scalability: verification time grows rapidly with model size and ε . The 45-second query on a $784 \rightarrow 32 \rightarrow 16 \rightarrow 10$ network illustrates that even small models can be expensive.
- One query per rival class: there is no built-in "argmax" constraint, requiring $n-1$ separate solves per sample.
- Windows support: Marabou must be built from source in a Linux environment (WSL); no Windows binary is available.
- Engine output: the C++ solver writes verbose logs directly to stdout via freopen, making it difficult to suppress without redirecting all Python output.